

Chapter 5 — Sequences & Mathematical Induction

Part 2: 5.2 & 5.3 Mathematical Induction

COMP 233 Discrete Mathematics | Birzeit University
Content follows Epp, Discrete Mathematics with Applications (4th ed.).

5.2 & 5.3 Induction - in this lecture

- 1** What induction is, and why it works
- 2 The method, and a first full proof
- 3 Sums: Gauss, arithmetic and geometric formulas
- 4 Divisibility and inequalities
- 5 Properties of recursive sequences
- 6 Induction vs. deduction, and more practice

A puzzle that needs a new kind of proof

The question

Using only 3-cent and 5-cent coins, which total amounts n can you pay exactly?

$$8 = 3 + 5, \quad 9 = 3 + 3 + 3, \quad 10 = 5 + 5, \quad 11 = 3 + 3 + 5$$

The pattern, and the problem

Every amount from 8 upward seems to work - but "8, 9, 10, 11, ... and onward forever" is infinitely many cases.

We cannot check them one by one. We need a method that proves a statement for all integers $n \geq 8$ at once. That method is mathematical induction.

The principle of mathematical induction

To prove $P(n)$ is true for all integers $n \geq a$

Basis step

(1) Basis: $P(a)$ is true

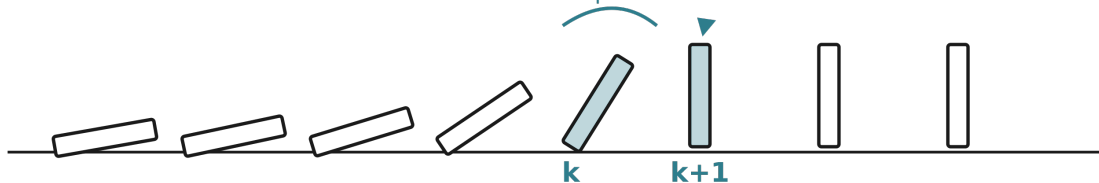
Inductive step

(2) Inductive step: $\forall k \geq a, P(k) \Rightarrow P(k+1)$

Conclusion

Then $P(n)$ is true for every integer $n \geq a$

If domino k falls, it knocks over domino $k+1$
pushes



Basis topples the first domino; the inductive step makes each falling domino topple the next.

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Anatomy of an induction proof

The three moving parts

1. **Basis step:** prove the statement directly at the starting value (often $n = 1$ or $n = 0$).
2. **Inductive hypothesis:** ASSUME the statement holds for one integer $k \geq a$.
3. **Inductive step:** USING that assumption, prove the statement for $k + 1$.

A common worry, settled

"Isn't assuming $P(k)$ the same as assuming what we want?" No. We assume it for a single k and show that forces $P(k+1)$. We never assume $P(n)$ for all n . The basis plus that single link is what makes the whole chain fall.

Worked proof: the coin problem

Claim (Epp, Proposition 5.2.1): every integer amount $n \geq 8$ can be paid with 3-cent and 5-cent coins.

Proof by induction on n

Basis step

$$P(8) : 8 = 3 + 5$$

Inductive step - assume some amount $k \geq 8$ is payable, and show $k + 1$ is too:

Case 1. the k -cent payment uses at least one 5 $\rightarrow$ swap one 5 for two 3s:

$$k - 5 + 6 = k + 1$$

Case 2. it uses only 3s (so at least three of them) $\rightarrow$ swap three 3s for two 5s:

$$k - 9 + 10 = k + 1$$

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Gauss and the power of a closed form

The story

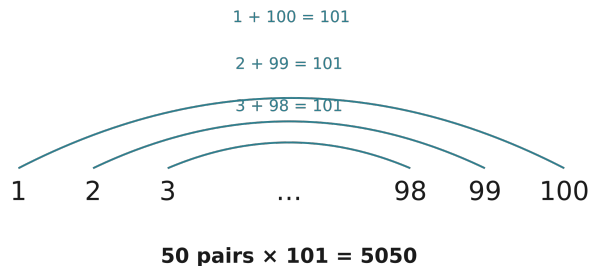
Asked to add $1 + 2 + \dots + 100$, the young Gauss paired the ends: $1 + 100$, $2 + 99$, $3 + 98$, ... Fifty pairs, each summing to 101.

$$1 + 2 + 3 + \dots + 100 = 5050$$

↑ *the sum*

↑ *the total*

A closed form gives the total directly - no loop, no adding term by term.



Theorem 5.2.2: sum of the first n integers

For every integer $n \geq 1$

$$1 + 2 + 3 + \dots + n = \frac{n(n+1)}{2}$$

↑ *the sum, written out*

↑ *the closed form*

The same thing, as a loop

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s = 0; for (k = 1; k ≤ n; k++) s = s + k;
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The loop adds the integers one at a time in n steps. The closed form $n(n+1)/2$ returns the same answer in a single step - and induction proves the two always agree.

Proof of Theorem 5.2.2 by induction

Induction on n

Basis step ($n = 1$)

$$n = 1 : 1 = \frac{1 \cdot 2}{2}$$

Inductive hypothesis (assume the formula at $n = k$)

$$1 + 2 + \dots + k = \frac{k(k+1)}{2}$$

Inductive step (add $(k+1)$ to both sides, then simplify)

$$\begin{aligned} 1 + 2 + \dots + k + (k+1) &= \frac{k(k+1)}{2} + (k+1) \\ &= \frac{(k+1)(k+2)}{2} \end{aligned}$$

Using the formula to evaluate sums

Sum of even numbers - factor the 2 out first

$$2 + 4 + \dots + 500 = 2 \cdot \frac{250 \cdot 251}{2} = 62750$$

Sum from 5 to 50 - whole sum minus the missing head

$$5 + 6 + \dots + 50 = \frac{50 \cdot 51}{2} - \frac{4 \cdot 5}{2} = 1265$$

Theorem 5.2.3: the geometric sum

For a fixed ratio $r \neq 1$ and integer $n \geq 0$

$$\sum_{i=0}^n r^i = \frac{r^{n+1} - 1}{r - 1} \quad (r \neq 1)$$

↑ *the geometric sum* ↑ *the closed form*

Why this is important

A geometric sum adds powers of a fixed ratio r : $1 + r + r^2 + \dots + r^n$. It appears everywhere - compound interest, loop-doubling costs, binary place values - and the closed form collapses the whole sum into one fraction.

Proof of the geometric sum, and an example

Induction on n

Basis step ($n = 0$)

$$n = 0: \quad r^0 = 1 = \frac{r^1 - 1}{r - 1}$$

Inductive hypothesis

$$\sum_{i=0}^k r^i = \frac{r^{k+1} - 1}{r - 1}$$

Inductive step (add the next term, combine over $r - 1$)

$$\sum_{i=0}^{k+1} r^i = \frac{r^{k+1} - 1}{r - 1} + r^{k+1} = \frac{r^{k+2} - 1}{r - 1}$$

Example (ratio $r = 2$)

$$1 + 2 + 2^2 + \cdots + 2^n = 2^{n+1} - 1$$

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Induction for divisibility (Proposition 5.3.1)

Claim

$$2^{2n} - 1 \text{ is divisible by } 3 \quad (n \geq 0)$$

Induction on n ("divisible by 3" means it equals $3 \times$ some integer)

Basis step

$$n = 0: \quad 2^0 - 1 = 0 = 3 \cdot 0$$

Inductive hypothesis (write the multiple of 3 explicitly as $3r$)

$$2^{2k} - 1 = 3r$$

Inductive step (peel out $4 \times$ the previous expression, leaving a multiple of 3)

$$2^{2(k+1)} - 1 = 4(2^{2k} - 1) + 3 = 3(4r + 1)$$

Induction for inequalities (Proposition 5.3.2)

Claim (note the start is $n = 3$, since it fails for smaller n)

$$2n + 1 < 2^n \quad (n \geq 3)$$

Induction on n

Basis step ($n = 3$)

$$n = 3: \quad 2(3) + 1 = 7 < 8 = 2^3$$

Inductive hypothesis

$$2k + 1 < 2^k$$

Inductive step (add 2 to the left; bound it using $2 \leq 2^k$ for $k \geq 3$)

$$2(k + 1) + 1 = (2k + 1) + 2 < 2^k + 2^k = 2^{k+1}$$

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Proving a closed form for a recursive sequence

A sequence defined by a recurrence

$$a_1 = 2, \quad a_k = 5 a_{k-1} \quad (k \geq 2) \quad \text{Claim: } a_n = 2 \cdot 5^{n-1}$$

Induction on n

Basis step ($n = 1$)

$$n = 1: \quad a_1 = 2 = 2 \cdot 5^0$$

Inductive hypothesis

$$a_k = 2 \cdot 5^{k-1}$$

Inductive step (use the recurrence, then the hypothesis)

$$a_{k+1} = 5 a_k = 5 \cdot 2 \cdot 5^{k-1} = 2 \cdot 5^k$$

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Inductive reasoning vs. deductive reasoning

Inductive reasoning

Starts from **observations and examples**.

Moves from **specific** → **general**.

Produces a **conjecture** - it suggests, it does not prove.

Deductive reasoning

Starts from **definitions and theorems**.

Moves from **general** → **specific**.

Produces a **proof** - the conclusion is guaranteed.

Watch the name: despite the word "induction", mathematical induction is a **deductive** method - it proves, with certainty, for all n .

More statements you can now prove by induction

Each follows the same recipe - basis, hypothesis, step. Sums and divisibility and inequalities, all in one toolkit:

Sum formula

$$\sum_{i=1}^n i(i+1) = \frac{n(n+1)(n+2)}{3}$$

Sum of cubes

$$\sum_{i=1}^n i^3 = \frac{n^2(n+1)^2}{4}$$

Divisibility

$8^n - 3^n$ is divisible by 5

Divisibility

$n^3 + 2n$ is divisible by 3

Inequality

$$n^2 < 2^n \quad (n \geq 5)$$

Recommended exercises (Epp, 4th ed.)

5.2 Model a problem like the coins / postage	1, 2
5.2 Prove a sum formula directly	6, 7, 8, 9
5.2 Prove the standard formulas ($\sum i^2$, $\sum i^3$, ...)	10, 11, 12, 13, 17
5.2 Evaluate a sum using the theorems	20, 21, 22, 23, 25, 26
5.3 Divisibility by induction	8, 9, 10, 11, 12, 14, 15
5.3 Inequalities by induction	16, 17, 18, 19, 20, 23
5.3 Closed form of a recursive sequence	24, 25, 26, 27

Summary

- Mathematical induction proves a statement for all integers $n \geq a$ using two steps: a basis case, and an inductive step $P(k) \Rightarrow P(k+1)$.
- The inductive hypothesis is the assumption $P(k)$ for one k ; you use it to force $P(k+1)$. It is not circular.
- Sum of the first n integers: $n(n+1)/2$. Geometric sum: $(r^{n+1} - 1)/(r - 1)$.
- The same method proves divisibility (write "divisible by d " as $d \times \text{integer}$) and inequalities (start the basis where the claim first holds).
- For a recursive sequence, the recurrence supplies the link the inductive step needs to prove a closed form.
- Despite its name, mathematical induction is a deductive method - it proves with certainty, for every n .